

AREA — Short Revision Notes

Evaluate Level: Make Judgments and Justify Decisions

1. Verifying Whether an Answer Is Correct

Problem	Answer	Verification	Judgment
Square side=5 cm	25 cm ²	$5 \times 5 = 25$	CORRECT ✓
Rectangle l=8, b=4	32 cm ²	$8 \times 4 = 32$	CORRECT ✓
Triangle base=10, height=6	60 cm ²	$\frac{1}{2} \times 10 \times 6 = 30$	INCORRECT ✗
Triangle base=10, height=6	30 cm ²	$\frac{1}{2} \times 10 \times 6 = 30$	CORRECT ✓
Parallelogram base=8, height=5	40 cm ²	$8 \times 5 = 40$	CORRECT ✓
Circle r=7 cm	154 cm ²	$22/7 \times 49 = 154$	CORRECT ✓
Circle r=7 cm	49 cm ²	Missing π	INCORRECT ✗

Method: identify the shape → apply the correct formula → calculate carefully → compare with the answer.

2. Judging the Correct Formula

Scenario	Formula Used	Correct?	Why / Why Not
Triangle area	base $\times$ height	WRONG ✗	Missing $\frac{1}{2}$
Triangle area	$\frac{1}{2} \times$ base $\times$ height	CORRECT ✓	Correct formula
Parallelogram area	$\frac{1}{2} \times$ base $\times$ height	WRONG ✗	No $\frac{1}{2}$ needed
Parallelogram area	base $\times$ height	CORRECT ✓	Correct formula
Trapezium area	$(a+b) \times h$	WRONG ✗	Missing $\frac{1}{2}$
Trapezium area	$\frac{1}{2} \times (a+b) \times h$	CORRECT ✓	Correct formula
Circle area	πd^2	WRONG ✗	Should use r^2
Circle area	πr^2	CORRECT ✓	Correct formula

Common Mistakes to Judge

Mistake	Why Wrong	Correct Version
Triangle: base×height	Missing $\frac{1}{2}$	$\frac{1}{2} \times \text{base} \times \text{height}$
Parallelogram: $\frac{1}{2} \times \text{base} \times \text{height}$	Has $\frac{1}{2}$ incorrectly	base × height
Trapezium: $(a+b) \times h$	Missing $\frac{1}{2}$	$\frac{1}{2} \times (a+b) \times h$
Circle: πd^2	Should use r^2	πr^2
Circle: πr^2	This is correct!	✓

3. Evaluating Reasonableness

Scenario	Answer	Reasonable?	Justification
Square side=10	100 cm ²	YES ✓	$10 \times 10 = 100$
Square side=10	1000 cm ²	NO ✗	Too large (=100)
Triangle b=10, h=5	25 cm ²	YES ✓	$\frac{1}{2} \times 10 \times 5 = 25$
Triangle b=10, h=5	50 cm ²	NO ✗	Missing $\frac{1}{2}$
Rectangle l=10, b=5	50 cm ²	YES ✓	$10 \times 5 = 50$
Rectangle l=10, b=5	30 cm ²	NO ✗	Too small
Circle r=7	154 cm ²	YES ✓	$\pi \times 49 = 154$
Circle r=7	49 cm ²	NO ✗	Missing π

Checklist: area > 0; triangle < rectangle (same b,h); parallelogram = rectangle (same b,h); circle $\approx 3.14 \times r^2$.

4. Judging the Best Shape for a Given Area

Scenario 1 — Which shape has more area?

Shape A	Shape B	Evaluation
Square side=10 (Area=100)	Rectangle l=12, b=8 (Area=96)	Square has more area (100 > 96)

Scenario 2 — Same perimeter, different areas

Shape	Measurements	Perimeter	Area	Ranking
Square	side=5	20	25	2nd
Rectangle	l=6, b=4	20	24	3rd
Circle	r≈3.18	20	≈31.8	1st

Judgment: for the same perimeter, the circle gives the largest area. For maximum area with minimum boundary, choose a circle.

5. Evaluating Student Mistakes

Student's Work	Analysis	Judgment	Correction
Triangle: $10 \times 6 = 60$	Forgot $\frac{1}{2}$	WRONG ✗	$\frac{1}{2} \times 10 \times 6 = 30$
Parallelogram: $\frac{1}{2} \times 8 \times 5 = 20$	Used $\frac{1}{2}$ incorrectly	WRONG ✗	$8 \times 5 = 40$
Trapezium: $(8+4) \times 5 = 60$	Forgot $\frac{1}{2}$	WRONG ✗	$\frac{1}{2} \times (8+4) \times 5 = 30$
Trapezium: $\frac{1}{2} \times (8+4) \times 5 = 30$	Correct formula	CORRECT ✓	✓
Circle: $\pi \times d^2 = 22/7 \times 14^2 = 616$	Used d instead of r	WRONG ✗	$r=7, \pi \times 7^2 = 154$

Common Error Types

Error Type	Example	Correct
Missing $\frac{1}{2}$ in triangle	$10 \times 6 = 60$	$\frac{1}{2} \times 10 \times 6 = 30$
Adding $\frac{1}{2}$ to parallelogram	$\frac{1}{2} \times 8 \times 5 = 20$	$8 \times 5 = 40$
Missing $\frac{1}{2}$ in trapezium	$(8+4) \times 5 = 60$	$\frac{1}{2} \times (8+4) \times 5 = 30$
Using diameter not radius	$\pi \times 14^2$	$\pi \times 7^2$
Radius not diameter (circumference)	$2 \times \pi \times r$	✓

6. Evaluating Compound Figure Solutions

Figure	Student's Work	Total	Evaluation
Square + Semicircle	$14 \times 14 = 196$, $\frac{1}{2} \times \pi \times 7^2 = 77$	$196 + 77 = 273$	CORRECT ✓
Square – Circle	$14 \times 14 = 196$, $\pi \times 7^2 = 154$	$196 - 154 = 42$	CORRECT ✓
Rectangle + Triangle	$12 \times 8 = 96$, $\frac{1}{2} \times 12 \times 8 = 48$	$96 + 48 = 144$	CORRECT ✓

Evaluating Mistakes

Figure	Student's Work	Evaluation	Why Wrong
Square + Semicircle	$14 \times 14 + \frac{1}{2} \times \pi \times 14^2$	WRONG ✗	Used diameter not radius
Square – Circle	$14 \times 14 - \pi \times 7^2$	CORRECT ✓	✓

7. Evaluating Real-World Problems

Scenario 1 — Painting a wall

Problem	Student's Work	Evaluation
Wall $10 \text{ m} \times 4 \text{ m}$, Door $2 \text{ m} \times 1.5 \text{ m}$; find area to paint	Wall = 40 m^2 , Door = 3 m^2 ; $40 - 3 = 37 \text{ m}^2$	CORRECT ✓

Scenario 2 — Tiling a floor

Problem	Student's Work	Evaluation
Floor $6 \text{ m} \times 5 \text{ m}$, Tile $30 \text{ cm} \times 30 \text{ cm}$; number of tiles	Floor = 30 m^2 , Tile = 0.09 m^2 ; $30 \div 0.09 \approx 334$	CORRECT ✓
Number of tiles	$30 \div 0.09 = 333.33 \rightarrow 334$ tiles	REASONABLE ✓

8. Comparing Two Solutions

Triangle: base 12 cm, height 8 cm

Solution A	Solution B	Evaluation
$\frac{1}{2} \times 12 \times 8 = 48 \text{ cm}^2$ (correct formula) — CORRECT ✓	$12 \times 8 = 96 \text{ cm}^2$ (forgot $\frac{1}{2}$) — WRONG ✗	

Parallelogram: base 10 cm, height 6 cm

Solution A	Solution B	Evaluation
$10 \times 6 = 60 \text{ cm}^2$ (correct formula) — CORRECT ✓	$\frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$ (used $\frac{1}{2}$ wrongly) — WRONG ✗	

9. Evaluating the Correct Base and Height

Figure	Student's Choice	Correct Choice	Evaluation
Triangle base 10, side 8	Height = 8	Height $\perp$ to base	WRONG ✗
Triangle base 10, height 6	Height = 6	Height = 6 ($\perp$)	CORRECT ✓
Parallelogram base 8, slant 5	Height = 5	Perpendicular distance	WRONG ✗

Justification: the height must be PERPENDICULAR to the base — not the slanted side.

10. Evaluating Perimeter vs Area Confusion

Student's Work	Analysis	Judgment
Square side=5, Area= $4 \times 5 = 20$	Used perimeter formula	WRONG ✗
Rectangle l=8, b=4, Area= $2(8+4) = 24$	Used perimeter formula	WRONG ✗
Triangle b=10, h=6, Area= $10+6+8 = 24$	Used perimeter formula	WRONG ✗

Correct areas: Square side=5 $\rightarrow 25 \text{ cm}^2$ (perimeter 20 cm); Rectangle l=8,b=4 $\rightarrow 32 \text{ cm}^2$ (perimeter 24 cm); Triangle b=10,h=6 $\rightarrow 30 \text{ cm}^2$.

11. Evaluating Unit Conversion

Problem	Student's Work	Evaluation
Rectangle $2\text{ m} \times 50\text{ cm}$	$2 \times 50 = 100\text{ cm}^2$	WRONG ✗
Rectangle $2\text{ m} \times 50\text{ cm}$	$200\text{ cm} \times 50\text{ cm} = 10,000\text{ cm}^2$	CORRECT ✓
Rectangle $2\text{ m} \times 0.5\text{ m} = 1\text{ m}^2$	$1\text{ m}^2 = 10,000\text{ cm}^2$	CORRECT ✓

Justification: units must be the SAME before multiplying.

12. Evaluation Checklist

When verifying an answer:

- ☐ Did I identify the shape correctly?
- ☐ Did I use the correct formula?
- ☐ Did I use the correct measurements?
- ☐ Did I calculate correctly?
- ☐ Is the answer reasonable?

When comparing options:

- ☐ Which shape gives more / less area?
- ☐ Which formula is correct?
- ☐ Which answer is more reasonable?

When judging student work:

- ☐ Correct formula? Correct base and height?
- ☐ Checked units? Showed working?

When evaluating real-world problems:

- ☐ Are the units consistent?
- ☐ Is the answer reasonable and complete?
- ☐ Does it make sense in context?

13. Self-Test — Evaluation Questions

No.	Question	Answer & Justification
1	Triangle b=10, h=6. Student says Area=60. Correct?	NO ✗ Should be 30 ($\frac{1}{2} \times 10 \times 6$)
2	Parallelogram b=8, h=5. Student says Area=40. Correct?	YES ✓ $8 \times 5 = 40$
3	Trapezium a=10, b=6, h=4. Student says Area=32. Correct?	YES ✓ $\frac{1}{2} \times (10+6) \times 4 = 32$
4	Circle r=7. Student says Area=154. Correct?	YES ✓ $\pi \times 49 = 154$
5	Circle d=14. Student says Area=154. Correct?	YES ✓ $r=7$, $\pi \times 49 = 154$
6	Square side=8. Student says Area=32. Correct?	NO ✗ Should be 64 (8×8)
7	More area: square side=6 or rectangle l=8, b=4?	Square: $36 > 32$
8	For same perimeter, which gives max area?	Circle
9	Why is triangle area less than rectangle (same b,h)?	Triangle is half of rectangle
10	Can parallelogram and rectangle have same area?	Yes, if same base and height

14. Quick Evaluation Guide

CORRECT if:

- Triangle uses $\frac{1}{2} \times \text{base} \times \text{height}$
- Parallelogram uses $\text{base} \times \text{height}$ (NO $\frac{1}{2}$)
- Trapezium uses $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$
- Circle uses πr^2 (NOT πd^2)
- Square uses a^2 ; Rectangle uses $l \times b$
- Height is PERPENDICULAR to base
- Units are the SAME before multiplying

INCORRECT if:

- Triangle uses $\text{base} \times \text{height}$ (missing $\frac{1}{2}$)
- Parallelogram uses $\frac{1}{2} \times \text{base} \times \text{height}$ (has $\frac{1}{2}$)
- Trapezium uses $(a+b) \times h$ (missing $\frac{1}{2}$)
- Circle uses πd^2 (should use r^2)
- Square uses $4a$ (that's perimeter)
- Rectangle uses $2(l+b)$ (that's perimeter)
- Slanted side used as height

15. Common Errors to Judge

Error	Example	Why Wrong	Correct
Forgetting $\frac{1}{2}$ in triangle	$10 \times 6 = 60$	Missing half	$\frac{1}{2} \times 10 \times 6 = 30$
Adding $\frac{1}{2}$ to parallelogram	$\frac{1}{2} \times 8 \times 5 = 20$	No $\frac{1}{2}$ used here	$8 \times 5 = 40$
Forgetting $\frac{1}{2}$ in trapezium	$(8+4) \times 5 = 60$	Missing half	$\frac{1}{2} \times (8+4) \times 5 = 30$
Using diameter in circle	$\pi \times 14^2 = 616$	Should use radius	$\pi \times 7^2 = 154$
Confusing area with perimeter	$4 \times 5 = 20$ for square	That's perimeter	$5 \times 5 = 25$

Revision Tip: when evaluating area problems, always check — correct formula? correct measurements (base, height, radius)? consistent units? reasonable answer? $\frac{1}{2}$ used correctly?